Public PSG to Polar H10

Measured algorithm development, independent testing and transfer analysis • 20-page report

Finding: the pilot learned a weak sleep-stage signal, but it did not transfer reliably enough for deployment.

This study used public sleep-study labels to develop a separate RR-only research model, then applied the frozen model to your real H10 recordings. It did not use demo recordings, D2 labels for training, or EEG-derived inputs. Your published OpenBeat application was left unchanged.

Measured result	Development: MIT	Independent test: UCD
Subjects / usable epochs	16 / 10,142	25 / 20,494
Four-stage accuracy	59.7%	37.0%
Macro F1 / Cohen kappa	0.412 / 0.307	0.318 / 0.086
Deep / REM recall	18.7% / 13.2%	23.4% / 14.1%

What this means for your recordings

Five real H10 sessions had raw RR intervals and could be processed; a sixth, cloud-only report could not. The two overnight outputs contained no predicted deep sleep and only 4 and 2 minutes of REM. These are model outputs, not findings about your actual sleep. Their mismatch with the app, poor external performance and signal-distribution differences argue against replacing the current heuristic.

Recommendation

Keep the present H10 staging experimental. Use this pipeline as a benchmark and a way to test candidate changes, with new public subjects held aside. Prioritize beat-processing consistency, relative rather than absolute heart features, missing-data robustness and larger, more varied PSG training data. None of these proposed improvements has yet been demonstrated by this pilot.

Reading guide: methods and internal evidence pp. 2-8; external validation pp. 9-11; H10 transfer pp. 12-17; recommendations and provenance pp. 18-20.

What data was actually used

Observed file counts and processing outputs, rather than planned dataset sizes

Role	Files / people	Data used
Development: MIT-BIH [1]	18 records / 16 people	Reviewed ECG beat times + PSG stages
Independent test: UCD [2]	25 records / 25 people	ECG from PSG .rec + stage text
Personal transfer	5 raw H10 sessions	Recorded RR intervals; no PSG labels

The MIT recordings contain 85.6 hours of complete 30-second frames; UCD contains 173.3 hours. These are waveform/frame durations, not total sleep time. The experiments scored only epochs with a recognized reference label and sufficient RR evidence.

Four-class label harmonization

Report class	MIT annotation	UCD stage code
Awake	W	0
Light	1 + 2	2 + 3 (N1 + N2)
Deep	3 + 4	4 + 5 (N3 + N4)
REM	R	1
Not scored	Movement / unknown	6 artifact / 7 unknown; absent labels

Both sources use legacy stage conventions. Merging N3 and N4 approximates the modern deep-sleep category; merging N1 and N2 intentionally discards their distinction. This study therefore evaluates four broad classes, not full five-stage clinical scoring.

These are clinically selected sleep-disordered-breathing datasets, not a representative healthy consumer sample. Their populations and recording conditions limit generalization. UCD is an independent source, but independence does not make its physiology or sensor processing equivalent to yours.

Sources [1-2] and attribution appear on p. 20. All 105 downloaded source files matched their official SHA256 checksums. Public raw data were processed locally.

How leakage was controlled

A prespecified experiment with subject separation and a frozen external test

Step	What was done
1. Define	Wrote PROTOCOL.md before outcomes; fixed features, model families, quality gate and selection rule.
2. Hold out people	Leave one MIT subject out at a time. slp01a/b stay together; slp02a/b stay together.
3. Fit within fold	Learned imputers, scaler, classifiers and transition counts use training subjects only.
4. Select	Highest pooled held-out macro F1, with balanced accuracy as tie-breaker.
5. Freeze and test	Refit selected family on all MIT subjects, then evaluate UCD without retuning.
6. Transfer	Apply the same frozen model to raw H10 intervals; assess outputs and robustness.

No target-stage labels, EEG signals, apnea-event labels, D2 labels or stage-selected baseline windows entered the predictors. Local relative features are computed from unlabelled signal context in each record. This is a fixed retrospective feature definition, separate from learned preprocessing fitted on training subjects.

Important limits of the design

The same internal cross-validation results selected the winner and describe its performance. They are not a nested, unbiased estimate of the selected model. The independent UCD result is the stronger check. Its poor result was retained rather than used to tune a more flattering model.

Quality gaps reset temporal decoding. Unknown reference labels are excluded from scoring and transition learning; at inference the decoder can pass through signal-eligible epochs without reference labels. This clarifies an implementation difference from the protocol wording about resetting at all excluded annotation stretches. No test labels were used to create decoding boundaries.

RR features and quality rules

The same feature extractor was applied to all three signal domains

Each prediction covers a 30-second frame. The extractor also uses centered 90-second and 300-second windows, plus unlabelled local context within approximately 45 minutes either side. This is an offline analysis: future observations are available. It is not a streaming model.

Feature family	Examples and purpose
Current frame	Mean heart rate, RR standard deviation, RMSSD, pNN50, accepted fraction, beat count and longest gap.
Contextual variation	90-second and five-minute HR / variability, RR quantiles, coefficient of variation and HR slope.
RR spectrum	LF and HF log power, LF fraction, HF peak frequency and spectral entropy. These are not direct respiration measurements.
Relative context	Local relative HR, relative RMSSD and HR variability; five-minute accepted-data fraction.

Missing observations were preserved

Intervals outside 300-2,000 ms, or more than 20% from a neighboring median, were rejected. An epoch needs at least 10 accepted beats and no gap over five seconds spanning its midpoint. These engineering thresholds were fixed before evaluation; they are not clinical definitions of a good ECG.

Successive-RR differences require consecutive accepted observations and consistent timing. Spectral resampling is at 4 Hz, with gaps above five seconds excluded from support and at least 75% support required. Interpolated samples never count as observed beats. Missing features are imputed within the trained pipeline, while failed epochs receive no stage.

There are 29 input features. Several encode related physiology, and feature availability itself can differ between datasets. The quality rule catches missing evidence; it cannot prove that a detected beat is correct or distinguish every artifact from arrhythmia.

Models and temporal criteria

This was a trained research companion, not a rewritten OpenBeat heuristic

Model	Fixed specification
Always-light baseline	Predict the training-majority class; exposes the effect of class imbalance.
Logistic regression	Balanced classes; C = 0.3; 1,000 iteration limit; median imputation, missing indicators and standard scaling.
Histogram boosting	Balanced classes; 150 iterations; learning rate 0.05; 15 leaves; minimum leaf 60; L2 = 2; seed 2026.
Temporal variants	First-order Viterbi decoding; training transition counts + one smoothing; log-transition weight 0.5; uniform initial state.

The temporal model favors plausible sequences learned from the training labels, while allowing the observations to overrule that preference. It does not impose a five-minute minimum stage, prohibit early REM, or assume that every recording begins at sleep onset. Decoding restarts at quality gaps.

What the model does not establish

A learned association between heart variability and a PSG label is not a physiological definition of deep sleep or REM. Those clinical stages depend on brain, eye and muscle activity. Cardiac evidence is indirect and can overlap across stages, quiet wake, illness and respiratory events.

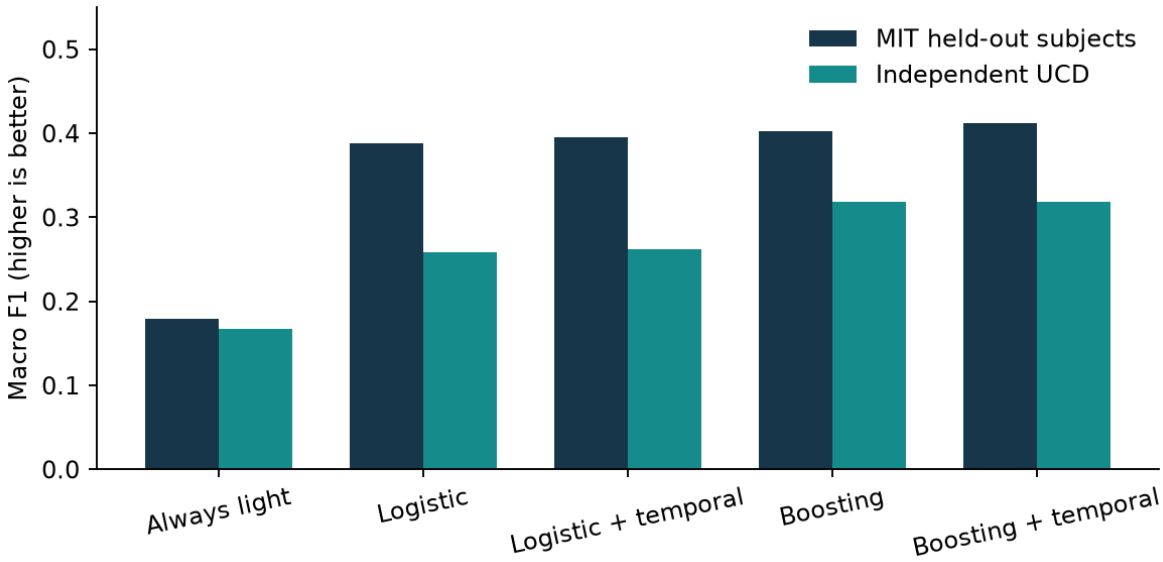
Class-balanced training helps expose minority stages, but the resulting output probabilities are not calibrated clinical confidence. The model is not an interpretable set of D2 thresholds. This experiment did not attempt to recover proprietary D2 firmware criteria.

The public datasets did not supply equivalent H10 chest acceleration. No artificial motion input was created. Consequently, this study cannot rank the current multisignal app against the new model on PSG accuracy; they use different evidence.

Internal results: all candidates

MIT leave-one-subject-out predictions • 10,142 usable reference epochs

Candidate	Accuracy	Bal. acc.	Macro F1	Kappa
Always light	56.1%	25.0%	0.180	0.000
Logistic	46.6%	46.8%	0.388	0.248
Logistic + temporal	48.1%	46.8%	0.395	0.257
Boosting	56.7%	40.9%	0.402	0.283
Boosting + temporal	59.7%	41.1%	0.412	0.307



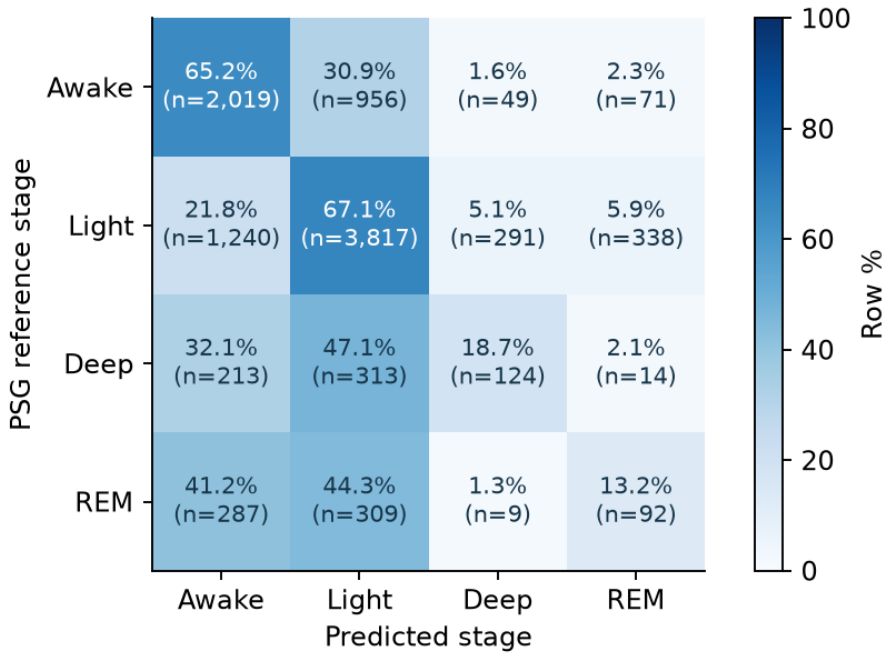
Boosting with temporal decoding won the prespecified macro-F1 selection. Its 59.7% accuracy is only 3.6 percentage points above always predicting light sleep (56.1%). Its better macro F1 and kappa show information beyond that baseline, but the gain is modest and does not imply useful performance for every stage.

Temporal decoding raised boosting accuracy from 56.7% to 59.7%, but macro F1 rose only from 0.402 to 0.412. This is evidence for a small sequence benefit in this dataset, not proof that a particular minimum stage duration is correct.

Balanced accuracy is the mean recall across the four classes. Macro F1 gives each class equal weight. Kappa adjusts observed agreement for agreement expected from the marginal label frequencies. External bars are shown here for context and discussed on pp. 9-10.

Which stages were recovered?

Internal selected-model confusion • rows are the PSG reference



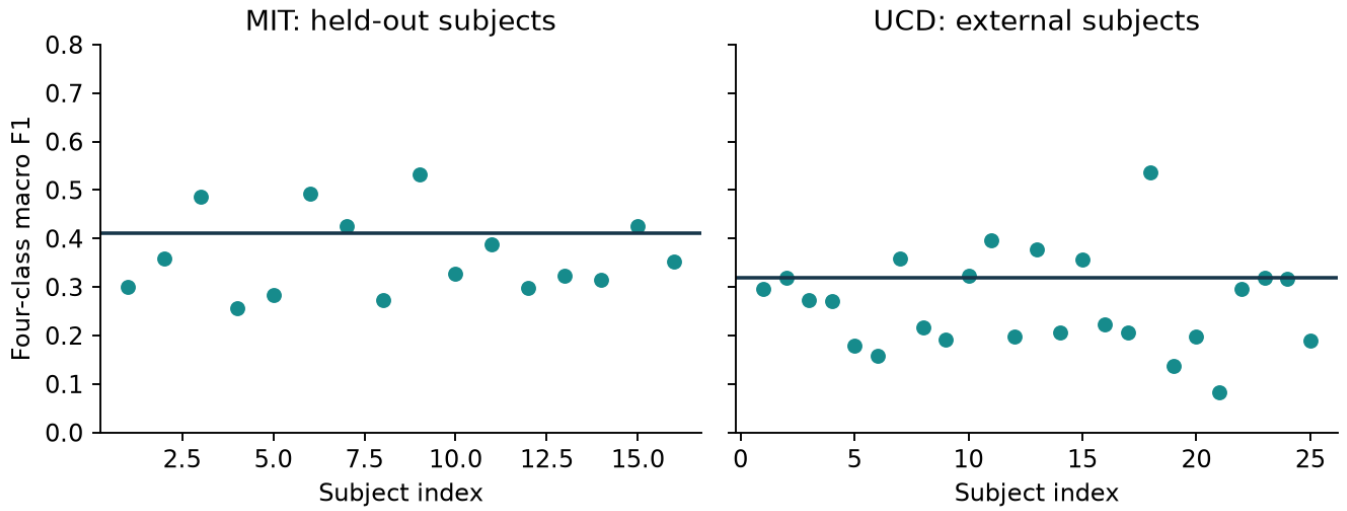
Stage	Precision	Recall	F1	Epochs
Awake	53.7%	65.2%	0.589	3,095
Light	70.8%	67.1%	0.689	5,686
Deep	26.2%	18.7%	0.218	664
REM	17.9%	13.2%	0.152	697

Awake and light dominate both the data and the useful performance. The model recovers only 124 of 664 deep epochs and 92 of 697 REM epochs. Many of these missed epochs become light or awake. Total accuracy therefore conceals poor stage sensitivity.

Deep and REM predictions are also unreliable when made: precision is 26.2% for deep and 17.9% for REM. A model with this error pattern should not be used to tell you that a particular low-heart-rate period was definitely deep sleep.

Variation and statistical uncertainty

People, not individual 30-second epochs, are the uncertainty units



Selected model	Pooled macro F1	Subject-bootstrap 95% interval
MIT held-out subjects	0.412	0.343-0.474
Independent UCD	0.318	0.278-0.354

The intervals resample complete subject confusion matrices 1,000 times, then recompute pooled four-class macro F1. They describe between-subject sampling variation under this experiment. They do not cover every source of uncertainty, such as label differences, detector errors or future populations.

Epochs from the same person are strongly related. Treating more than 30,000 epochs as independent trials would make the uncertainty appear much smaller than justified. Sixteen development people remain a small training cohort despite the many observations.

The chart uses a fixed four-class definition for each person; a class absent from a person contributes zero F1. Thus these subject scores are descriptive and not directly interchangeable with studies that omit absent classes. Internal selection also creates optimism not removed by this bootstrap.

Independent external test

UCD PSG ECG • 25 people • 20,494 usable reference epochs

Frozen candidate	Accuracy	Bal. acc.	Macro F1	Kappa
Always light	50.5%	25.0%	0.168	0.000
Logistic	25.8%	34.5%	0.259	0.079
Logistic + temporal	26.3%	34.1%	0.262	0.078
Boosting	36.2%	33.0%	0.318	0.088
Boosting + temporal	37.0%	32.8%	0.318	0.086

The selected model lost 22.7 percentage points of accuracy outside its development source.

The selected boosting-temporal model achieved 37.0% accuracy, 32.8% balanced accuracy, macro F1 0.318 and kappa 0.086. Always predicting light achieved 50.5% accuracy. The trained model is therefore worse on raw accuracy than this simple baseline, although its macro F1 is higher because it predicts minority classes.

This distinction matters: the result is neither “no signal at all” nor “successful transfer.” Balanced accuracy is above the 25% equal-recall baseline, but class recovery remains weak and the errors are too substantial for deployment.

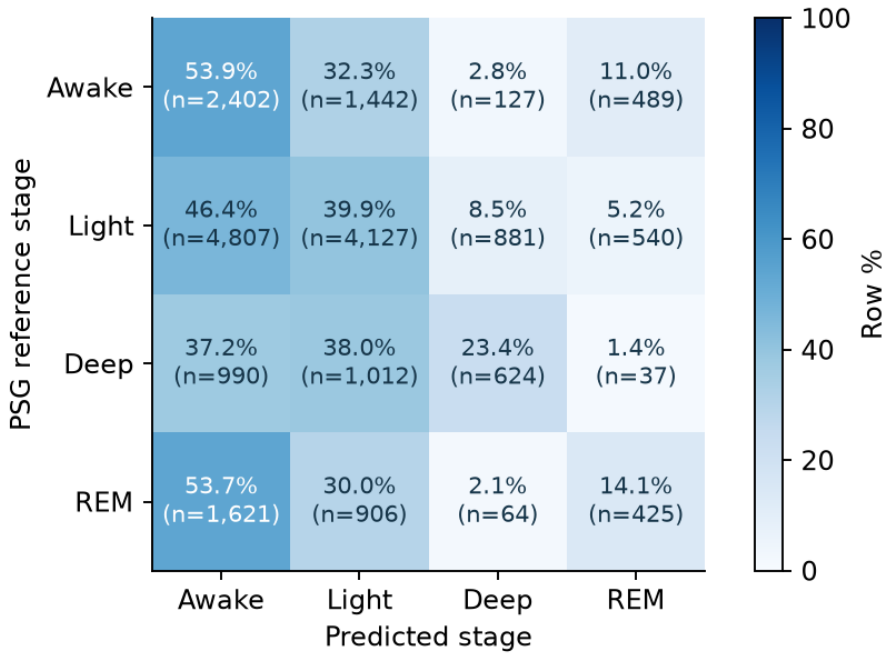
The negative result was not tuned away

All prespecified candidates are reported, and the internally selected model remained the primary external model. UCD outcomes did not trigger new thresholds, parameter searches, stage-duration adjustments or a switch to whichever external candidate happened to look best.

UCD is now a consumed evaluation set for this study. If it informs the next design, it should be treated as development evidence; a new untouched dataset is needed for the next strong external claim.

External errors: wake is overcalled

Independent selected-model confusion • rows are the PSG reference



Stage	Precision	Recall	F1	Epochs
Awake	24.5%	53.9%	0.336	4,460
Light	55.1%	39.9%	0.463	10,355
Deep	36.8%	23.4%	0.286	2,663
REM	28.5%	14.1%	0.189	3,016

Of 10,355 light epochs, 4,807 were classified awake. Of 3,016 REM epochs, 1,621 were classified awake. This is a pronounced wake-overcalling failure, not simply an exchange between neighboring sleep stages.

Deep recall is 23.4% and REM recall 14.1%. The slightly better deep recall than on MIT does not rescue the model: the overall pattern changes, wake precision is only 24.5%, and the independent kappa remains weak. A plausible-looking hypnogram would not override these measured errors.

Signal quality and processing differences

Availability is measurable; detector correctness was not independently adjudicated

Dataset	Recognized labels	Usable labels	Usable / recognized
MIT	10,181	10,142	99.62%
UCD	20,774	20,494	98.65%

MIT used reviewed beat annotations. UCD used automatically detected beats from the ECG channel inside the PSG file, with the WFDB XQRS detector. This avoided the separate Holter file whose timing could not be assumed to match the stage labels. It also introduced a real processing-domain difference between development and external testing.

Post-hoc continuity check: descriptive only

Source / stratum	Epochs	Accuracy	Macro F1
MIT / high continuity	8,991	59.3%	0.409
MIT / other eligible	1,151	62.2%	0.351
UCD / high continuity	18,175	35.7%	0.315
UCD / other eligible	2,319	47.2%	0.343

High continuity means at least 95% accepted observed intervals in the target frame and no observed accepted-beat gap over two seconds. Other eligible epochs still pass the main gate. This post-hoc stratification did not refit or select any model, and differences also reflect stage and subject composition; they are not causal estimates of artifact effects.

High coverage cannot certify accurate QRS detection, breathing estimates or stages. Population, scorer conventions, respiration pathology and beat-processing differences are confounded here. A useful follow-up is to process development and test ECG with the same detector and manually audit a sample of beats before attributing the failure to physiology alone.

Which personal recordings transferred?

Your backup only • Singapore time • demo sessions excluded

Start	Whole frames	RR accepted	Epochs eligible
04 Nov 07:40	109.0 min	91.6%	95.9%
04 Nov 10:05	13.0 min	90.6%	96.2%
04 Nov 10:57	56.5 min	57.4%	58.4%
04 Nov 23:31	419.0 min	86.0%	89.1%
06 Nov 01:53	279.0 min	84.1%	85.5%

These five sessions contain real interval observations. A sixth session, beginning 6 November at 15:56 and lasting about 116.6 minutes, is a cloud-only copy without raw RR. It was excluded from model transfer rather than reconstructed from summary values.

All durations and tables use complete 30-second frames; leftover fractions of a minute are not predicted. “RR accepted” is the fraction of supplied intervals surviving cleaning. “Epochs eligible” is the fraction passing the research model’s frame gate. Neither is the app’s multisignal coverage percentage.

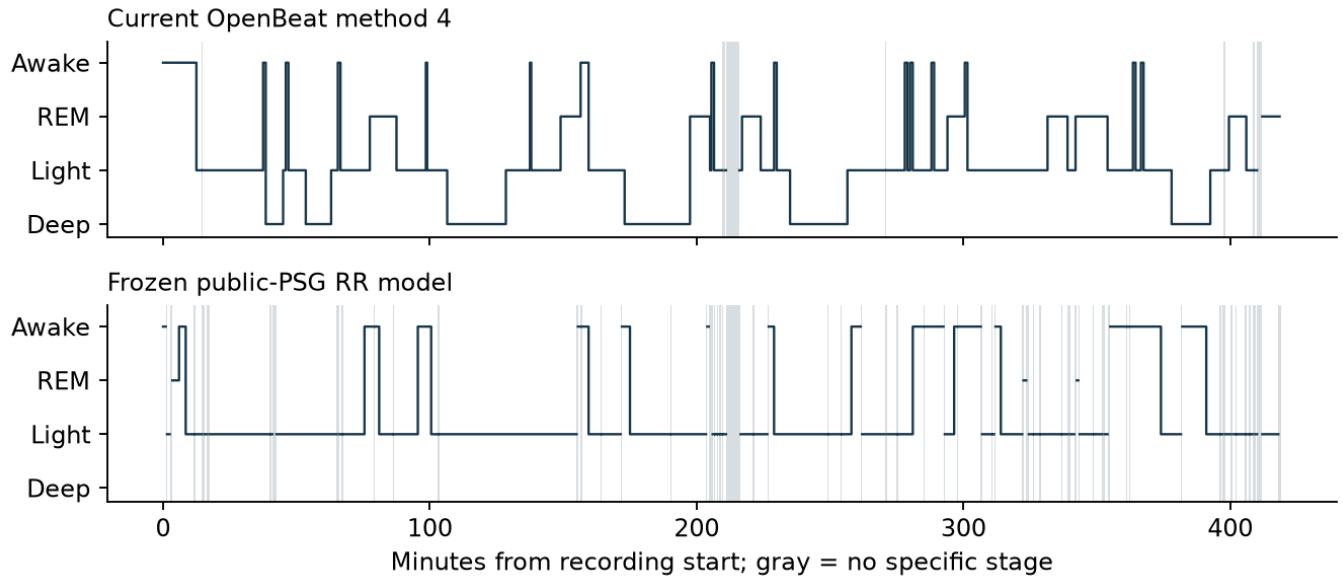
What can and cannot be measured here

We can measure missingness, distribution shifts, predicted stage totals, sensitivity to artificial perturbations and agreement with the current app. We cannot measure your H10 stage accuracy because these sessions have no simultaneous PSG labels. D2 agreement would be another device comparison, not a substitute for clinical truth.

The current OpenBeat method-4 algorithm was rerun from the supplied v1.34.1 snapshot, using default onset settings, as a descriptive comparator; saved per-report overrides were not imported. Its stage output was not used to train, choose or calibrate the research model. Daytime recordings are identified by time; actual sleep during every minute was not independently verified.

Overnight: 4-5 November

Recording starts 4 November 23:31 • 419.0 complete minutes



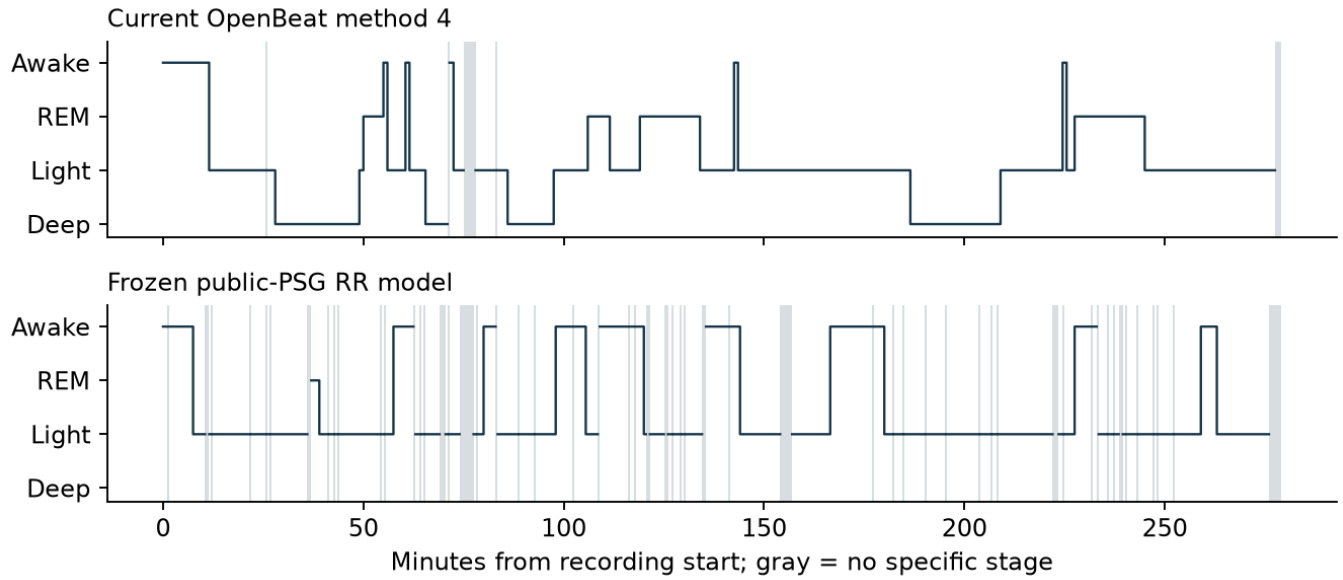
Full-record minutes	Current app	Public-PSG model
Awake	27.5	75.5
Light	211.5	294.0
Deep	98.5	0.0
Rem	71.5	4.0
No specific stage	10.0	45.5

Agreement on the 373.0 minutes where both methods give one of four stages is 42.0%. This is conditional algorithm agreement, not accuracy. The model labels no deep sleep and just 4 minutes of REM, despite the app assigning 98.5 and 71.5 minutes respectively.

Gray intervals differ by method because their evidence and quality gates differ. App awake minutes here include pre-onset wake; they therefore differ from its within-sleep awakening metric. Without PSG, neither line settles the true sleep architecture. The external failure gives a positive reason to distrust the new model’s extreme stage totals.

Overnight: 6 November

Recording starts 01:53 • 279.0 complete minutes • partial-night context



Full-record minutes	Current app	Public-PSG model
Awake	16.5	62.0
Light	153.0	174.5
Deep	60.5	0.0
Rem	43.0	2.0
No specific stage	6.0	40.5

Both methods stage 238.0 common minutes, with 42.4% agreement. The new model again predicts zero deep sleep and only 2 minutes of REM. Its accepted-epoch coverage is 85.5%, leaving 40.5 minutes without a stage.

Starting late in the night changes the physiological context available to an algorithm. This model uses neither clock time nor a hard early-REM exclusion, yet it still fails to produce convincing transfer. Removing a timing rule alone is therefore insufficient evidence of a better classifier. This single recording cannot identify the correct late-start adjustment.

The three daytime recordings

Short recordings and missing intervals stress the same frozen model

Start, 4 Nov	Awake	Light	Deep	REM	No stage
07:40	76.0	17.0	4.5	7.0	4.5
10:05	3.0	9.5	0.0	0.0	0.5
10:57	12.5	16.5	0.0	4.0	23.5

All stage values are minutes, rounded to complete half-minute frames. “No stage” is failed research quality eligibility, not confirmed sleep or wake.

07:40: substantial disagreement

The research model predicts 76 minutes awake and only 28.5 minutes of classified sleep in the 109-minute frame span. The current app reports 99.5 minutes of sleep. This disagreement does not establish that you were awake; it demonstrates that the new cardiac model is not interchangeable with the app.

10:05: a very short record

The 13-minute span has 9.5 minutes light, 3 minutes awake and half a minute without a model stage. Five-minute centered windows are truncated near both ends, so there is much less independent contextual information than in a full night. No inference about habitual nap staging follows from this example.

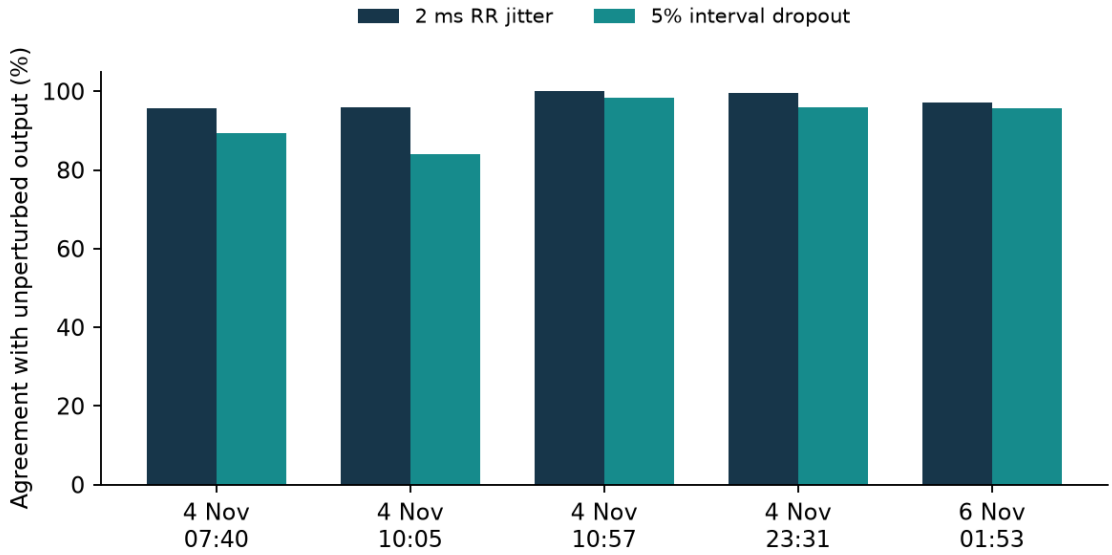
10:57: missingness dominates

Only 58.4% of frames pass the research gate; 23.5 minutes remain unclassified. The current app reports 13.5 minutes of uncertain sleep and higher coverage under its different evidence rules. That difference is expected from the inputs and thresholds; it is not proof that one missing-data definition is correct.

These records are useful technical stress tests, but they are not a labelled nap-validation dataset. Public overnight PSG results cannot be advertised as measured nap accuracy.

Robustness is not accuracy

Synthetic perturbations of the same personal intervals • seed fixed at 2026



Start	2 ms jitter agree	5% dropout agree	Dropout eligible
04 Nov 07:40	95.7%	89.4%	95.0%
04 Nov 10:05	96.0%	84.0%	96.2%
04 Nov 10:57	100.0%	98.5%	58.4%
04 Nov 23:31	99.6%	96.0%	88.9%
06 Nov 01:53	97.1%	95.8%	85.3%

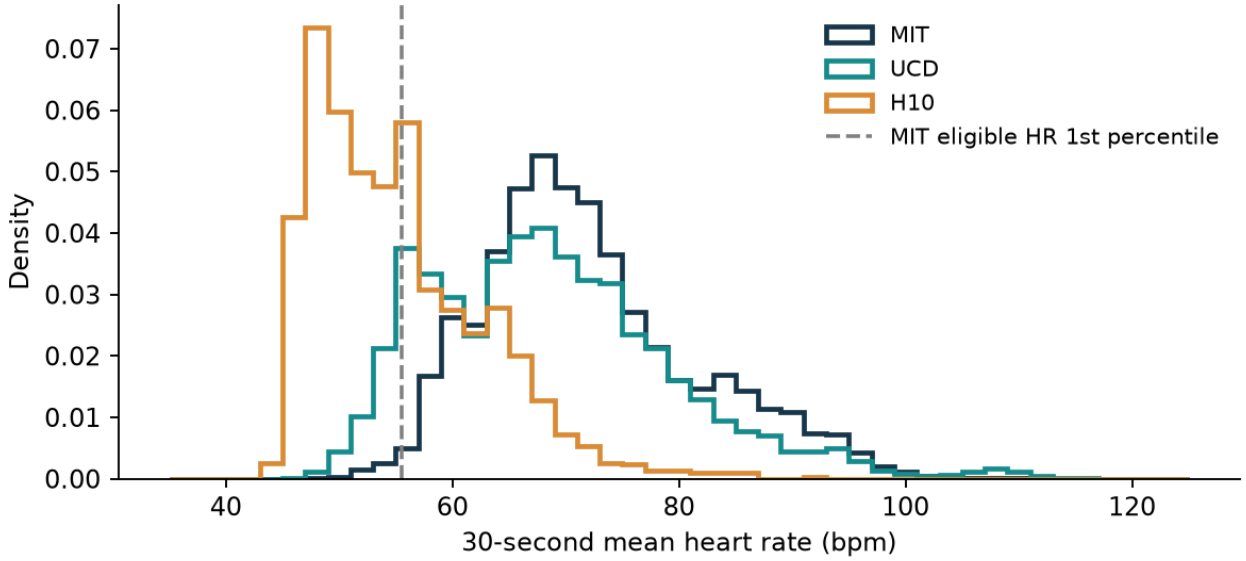
The first perturbation adds Gaussian timing noise with 2 ms standard deviation to RR values. The second removes a random 5% of supplied interval observations without inventing replacements. Features and quality gates are recomputed, then the same model is applied. This is one seeded stress test per record, not a sensitivity confidence interval.

Agreement is calculated only where both original and perturbed outputs are eligible. The accompanying coverage column prevents removed evidence from disappearing silently. Conditional agreement may look high partly because difficult frames no longer qualify.

The overnight outputs are fairly stable under these small changes. A stable error remains an error: this result cannot compensate for poor PSG performance or establish that the model’s near-absence of deep and REM sleep is correct.

Why transfer is especially doubtful

The personal signal distribution is outside much of the development experience



Eligible HR30	MIT	UCD	Your H10
10th percentile	60.9	55.7	47.2
Median	70.5	67.7	53.9
90th percentile	86.4	82.8	65.6
Finite feature cells outside training 1st-99th	1.8%	9.0%	17.2%

Across eligible personal frames, 57.5% have heart rate below the labelled-training 1st-percentile boundary. Around one fifth of finite feature cells are outside the training envelope on each overnight record. Absolute heart-rate thresholds and tree splits may therefore behave differently on your nights.

The envelope is a marginal screening diagnostic, not a multivariate out-of-distribution probability or a medical abnormality threshold. Signal quality, physiology and sampling context can all shift these distributions. Lower personal heart rate by itself does not prove better fitness, deeper sleep, or detector failure.

Changes I recommend next

Research priorities for approval before any production heuristic changes

Priority	Candidate experiment	Acceptance evidence
1	Use the same ECG detector across public sources; audit a sample of beats and timestamps.	External errors cannot be explained by a basic alignment or beat-processing mistake.
2	Ablate absolute HR / RR level features; test relative and person-normalized alternatives.	Improvement on untouched subjects, including low-HR strata; no use of D2 as truth.
3	Train on larger, more diverse PSG ECG cohorts and multiple sources.	Independent four-class macro F1, per-stage recall and subject uncertainty improve.
4	Add realistic missingness / artifact stress tests and evaluate abstention.	Coverage is reported beside performance; abstention does not conceal failures.
5	Evaluate sequence decoding separately from evidence scoring.	Better per-stage results without imposing attractive but false stage durations.

For the published H10 heuristic, keep explicit missingness, separate modality coverage, experimental labelling and version provenance. Preserve the current five-minute rule until it is tested directly; this pilot did not establish a better duration rule. Keep quiet-wake suspicion as an annotation rather than an unvalidated stage override.

Do not copy the pilot’s thresholds, force its stage totals toward D2, or use normal population stage percentages as individual targets. Its external test is too weak. The immediate recommendation is a research plan, not a new set of numerical clinical cutoffs.

No source code changes are included in this report. Candidate experiments can be implemented in the isolated pipeline first, with their results presented for your approval before altering the app.

A practical path without your own PSG

What public labels can support, and what remains unverified

Next study: public-to-public generalization

Expand to another PSG ECG resource such as the Haaglanden Medisch Centrum sleep-staging database [3], after checking access, scoring, synchronization and license. This resource was investigated as a future option but was not downloaded, trained on or evaluated in this pilot. Reserve a fresh source or prospective subject split before tuning.

Define success in advance: four-class macro F1 and balanced accuracy, per-stage sensitivity and precision, usable coverage, and subject-level intervals. Compare against always-light and simple cardiac baselines. Report subgroup and poor-quality failures rather than publishing only a pooled accuracy.

Then assess transfer to H10 as a separate question

On additional real H10 nights, examine beat acceptance, artifact handling, feature distributions, repeatability and changes across recording lengths. Bedtime notes and a sleep diary can help check broad sleep windows and known wake periods. They cannot label N2, deep sleep or REM reliably.

D2 may remain an independent consumer comparator, with synchronized epochs and explicit missingness. Agreement can diagnose discrepancies but cannot decide which device is right. Training to mimic D2 would produce a D2-surrogate model, not PSG validation.

What would eventually justify a stronger claim?

A model that performs well on genuinely independent public PSG still needs simultaneous reference testing on the intended H10 setup to quantify its personal sensor transfer. Until such data are available, describe H10 stages as experimental estimates. Your lack of PSG access does not prevent useful development; it limits the claims that development can support.

The current failure is informative: public labels alone do not guarantee transfer. Better beat processing, representative training data and disciplined external testing should precede deployment or more elaborate stage rules.

Sources, audit trail and deliverables

Reproducible local analysis • no personal recordings uploaded

[1] Development data. Moody G. MIT-BIH Polysomnographic Database, v1.0.0. PhysioNet, 1999. doi:10.13026/C23K5S. Original publication: Ichimaru Y, Moody GB. Psychiatry Clin Neurosci. 1999;53:175-177. Reviewed ECG annotations and stage labels were used; raw MIT waveforms were not needed for this pilot.

[2] External data. St. Vincent's University Hospital / University College Dublin Sleep Apnea Database, v1.0.0. PhysioNet, 2007. doi:10.13026/C26C7D. Contributions: McNicholas, Doherty, Ryan, Garvey, Boyle and Chua. Used PSG ECG and associated stage labels. Both public datasets are under Open Data Commons Attribution License v1.0.

[3] Future dataset, not evaluated. Haaglanden Medisch Centrum sleep-staging database, v1.1. PhysioNet resource, doi:10.13026/t79q-fr32. Listed as a candidate expansion, not evidence for this study's performance.

[4] Platform attribution. Pollard T et al. PhysioNet as a global platform for biomedical research. Nature Health. 2026. doi:10.1038/s44360-026-00096-z. Dataset pages and documentation checked 6 November 2026.

Audit and reproducibility

The companion folder contains the protocol, extraction and model scripts, fixed model, metric JSON, epoch predictions, figures and download instructions. The distributable ZIP excludes raw public waveforms, dependency binaries and the private backup. It includes personal derived stage outputs; keep it private if you do not wish to share these.

Input: openbeat-backup-2026-11-06-10-57.json. App comparator: supplied "2026-11-06 2048 publish" snapshot, v1.34.1 / sleep method 4. Model SHA256: 336dcc3bd5ab43faa4aa0f65f1c98e7c8a324c923a0de2d0cc0f639461540acb. Full input and source hashes are in the result manifests.

Validation checks passed for 18 MIT records / 16 subject groups, 25 UCD records, five raw H10 sessions, all 105 source checksums, label padding, frame alignment, feature dimensions, grouping, metric recomputation and a constant-RR extraction case. An annotation-padding issue in two MIT records was corrected before all final internal results were rerun.

Decision: retain the current experimental app; do not deploy this pilot model. Use its measured failures to guide the next public-PSG experiment.